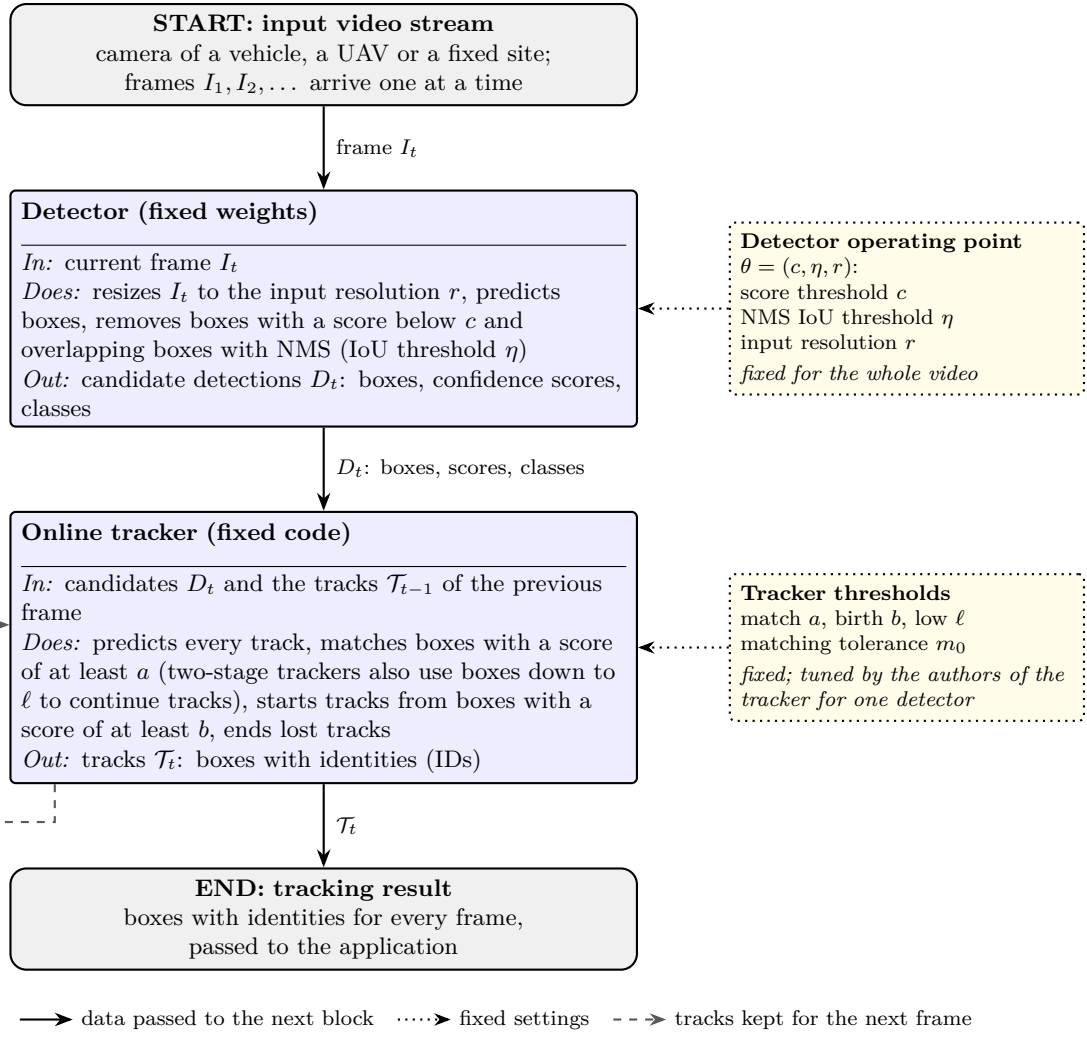

START: input video stream
camera of a vehicle, a UAV or a fixed site;
frames $I_1, I_2, \ldots$ arrive one at a time
frame I_t
Detector (fixed weights)
In: current frame I_t
Does: resizes I_t to the input resolution r, predicts boxes, removes boxes with a score below c and overlapping boxes with NMS (IoU threshold η)
Out: candidate detections D_t: boxes, confidence scores, classes
Detector operating point
$\theta = (c, \eta, r)$:
score threshold c
NMS IoU threshold η
input resolution r
fixed for the whole video
D_t: boxes, scores, classes
Online tracker (fixed code)
In: candidates D_t and the tracks $\mathcal{T}_{t-1}$ of the previous frame
Does: predicts every track, matches boxes with a score of at least a (two-stage trackers also use boxes down to ℓ to continue tracks), starts tracks from boxes with a score of at least b, ends lost tracks
Out: tracks $\mathcal{T}_t$: boxes with identities (IDs)
$\mathcal{T}_{t-1}$ for frame $t+1$
Tracker thresholds
match a, birth b, low ℓ
matching tolerance m_0
fixed; tuned by the authors of the tracker for one detector
$\mathcal{T}_t$
END: tracking result
boxes with identities for every frame,
passed to the application
data passed to the next block
fixed settings
tracks kept for the next frame